

11.14 Classwork: Calculus practice (no calculator)

Part A. Differentiate. Use the chain rule to find $\frac{dy}{dx}$.

1. $y = (3x^2 + 1)^4$

2. $y = \sqrt{2x^2 + 1}$

3. $y = \sin(2x^3)$

4. $y = e^{x^2 - 4x}$

5. $y = \ln(5x^2 + 3)$

6. $y = \cos^3(2x)$

Part B. Integrate. For each, use the reverse chain rule (substitution) to find $f(x)$.

7. Given $f'(x) = \frac{6x}{\sqrt{x^2 + 4}}$ and $f(0) = 5$, find $f(x)$.

8. Given $f'(x) = 2x \cos(x^2)$ and $f(0) = 1$, find $f(x)$.

9. Given $f'(x) = \frac{2x}{x^2 + 1}$ and $f(1) = 0$, find $f(x)$.

10. Given $f'(x) = \frac{\ln x}{x}$ and $f(1) = 3$, find $f(x)$.